CAB315:ARTIFICIAL INTELLIGENCE ANALYST 3

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: explain the historical evolution, paradigms, foundational concepts, limitations, and transitions of artificial intelligence and their relevance to modern intelligent systems.

CO2 :: apply state-space search, heuristic methods, logic, and rule-based reasoning to solve well-defined artificial intelligence problems.

CO3 :: evaluate neural-network learning mechanisms and deep learning architectures, including CNNs, RNNs, LSTMs, and GRUs, for vision, sequence, and temporal applications.

CO4 :: develop an AI solution using TensorFlow or PyTorch and appropriate Generative AI techniques while considering deployment, performance, ethics, safety, security, copyright, and enterprise requirements.

Unit I

HISTORY AND EVOLUTION OF ARTIFICIAL INTELLIGENCE - PARADIGMS, LIMITATIONS, AND TRANSITIONS : Understand the philosophical and computational origins of artificial intelligence, study the evolution of AI from symbolic reasoning and logic-based systems to learning-based approaches, examine key milestones and contributors in AI history, understand classical AI paradigms and their limitations, analyze the causes and impact of AI winters, understand how traditional AI concepts laid the foundation for modern deep learning systems

Unit II

CLASSICAL AI TECHNIQUES - SEARCH, REASONING, AND RULE-BASED INTELLIGENCE : Understand problem formulation using state-space representation, study uninformed search techniques, including breadth-first search and depth-first search, analyze informed search methods,, heuristic search, best-first search, A* algorithm, Dijkstra's algorithm, understand heuristic design and evaluation, study knowledge representation using propositional and predicate logic, explore rule-based systems, inference mechanisms, forward and backward chaining, analyze expert systems as classical AI problem-solving frameworks.

Unit III

ARTIFICIAL NEURAL NETWORKS – ARCHITECTURES, LEARNING AND OPTIMIZATION : Understand the biological inspiration and mathematical foundations of artificial neural networks, perceptron and multi-layer feed-forward neural networks, activation functions and loss functions, backpropagation, gradient-based optimization conceptually, vanishing and exploding gradients, regularization and generalization in deep networks, explore deep learning model-design considerations.

Unit IV

DEEP LEARNING MODELS – CONVOLUTIONAL AND RECURRENT NEURAL NETWORKS : convolutional neural networks, including convolution, pooling, and hierarchical feature extraction, analyze CNN architectures, applications in vision and pattern recognition, recurrent neural networks for sequence modeling,, long short-term memory networks, gated recurrent units, analyze temporal dependencies and memory mechanisms, real-world applications of CNNs, RNNs, LSTMs and GRUs in speech, language, vision and time-series analysis.

Unit V

DEEP LEARNING USING TENSORFLOW AND PYTORCH - FRAMEWORKS, WORKFLOWS, AND APPLICATIONS : architecture and design philosophy of TensorFlow and PyTorch, tensors, computational graphs and automatic differentiation, model definition, training and evaluation workflows in both frameworks, analyze differences between static and dynamic computation graphs, understand GPU acceleration and performance optimization conceptually, deployment considerations, and analyze ethical concerns, limitations and real-world applications of deep learning systems.

Unit VI

GENERATIVE ARTIFICIAL INTELLIGENCE : Introduction to Generative AI, evolution of foundation models, Large Language Models (LLMs), overview of GPT and Gemini, prompt engineering techniques, text, image, and code generation, AI assistants and copilots, Retrieval-Augmented Generation (RAG), AI agents, LangChain fundamentals, AI workflow automation, fine-tuning concepts, responsible Generative AI, hallucination and prompt injection, AI safety, copyright and intellectual property, enterprise use cases of Generative AI, future trends in Generative AI.

List of Practicals / Experiments:

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- Implement Breadth-First Search and Depth-First Search.
- Create an AI agent to solve the 8-puzzle problem using A* algorithm.
- Build a chatbot using Natural Language Toolkit (NLTK).
- Develop a rule-based expert system for medical diagnosis.
- Train a neural network for image classification using a small dataset.
- Implement a mini decision tree for predicting student performance.
- Create a fuzzy logic-based temperature control system.
- Design a basic reinforcement learning simulation (Grid World).
- Analyze a real-world dataset and apply supervised learning algorithms.
- Build a mini project showcasing AI application in any domain.

Text Books:

1. ARTIFICAL INTELLIGENCE: STRUCTURES AND STRATEGIES FOR COMPLEX PROBLEM SOLVING by GEORGE F. LUGER, PEARSON

2. ARTIFICIAL INTELLIGENCE: A MODERN APPROACH by RUSSELL, NORVIG, PEARSON

References:

1. ARTIFICIAL INTELLIGENCE by SAROJ KAUSHIK, Cengage India Private Limited